

Process Characterization Plan

**A-Mab Drug Substance — Small-Virus Retentive Filtration (Step 9) —
PCP-009**

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion exchange chromatography; ALCOA+, attributable, legible, contemporaneous, original and accurate, plus complete, consistent, enduring and available; AMV, analytical method validation; BLA, Biologics License Application; CPP, critical process

parameter; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance; ELN, electronic laboratory notebook; GPP, general process parameter; ICH, International Council for Harmonisation; KPP, key process parameter; LIMS, laboratory information management system; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; PD, Process Development; QA, Quality Assurance; qPCR, quantitative polymerase chain reaction; RSM, response surface methodology; SDM, scale-down model; SOP, standard operating procedure; TCID₅₀, fifty percent tissue culture infective dose; WC-CPP, well controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

The A-Mab drug substance process is being transferred from Novacyte Biologics — Cambridge, MA (Development) to Novacyte Biologics — Grafton, WI (Commercial DS). Stage 1 process characterization supports that transfer by establishing which process parameters affect product quality and by defining the ranges within which the commercial process may be operated (U.S. Food and Drug Administration 2011). This plan describes the characterization study for Step 9 of the drug substance train, the small virus retentive filtration step, and it is issued before any characterization data have been generated. The commercial process runs at a production bioreactor scale of 15,000 L.

The step is the dedicated small virus removal operation of the purification train and the principal source of MVM clearance. Its clearance is orthogonal to the two other viral safety steps, low pH inactivation (Step 6) and anion exchange chromatography (Step 8), because it removes virus by size rather than by inactivation or by binding (International Council for Harmonisation 2023a). The study covers the 2 process parameters of the step and the 3 responses listed in Table 7.1. It will be executed on a qualified scale-down model, on load material representative of the anion exchange pool that feeds the step at commercial scale.

Three things are outside the scope of this plan. Qualification of the membrane itself is held by the filter manufacturer and is not repeated here. The cumulative viral clearance claim for the drug substance is consolidated across all three viral safety steps in PCMR-001, and this study establishes only the contribution of Step 9. Process validation at commercial scale is covered by the validation programme and not by this study.

The deliverable of the study is PCR-009, the process characterization report for this step.

2 Objectives

The study has the following objectives.

1. Qualify a scale-down model of the small virus retentive filtration step, and show that it represents the commercial step for virus retention, for flux decay and for step yield.
2. Determine which of the parameters in Table 6.1 affect the two viral clearance responses and the step yield, in what direction and by how much.
3. Fit a response-surface model for each response over the characterization ranges, and establish whether it predicts the mean response adequately.
4. Derive a proven acceptable range for each parameter against each of the two viral clearance responses, using the criteria in §7 and the method in §8.
5. Classify each parameter under SOP-4001 and define what the step contributes to the control strategy and to the modular viral clearance claim.

The screening design identifies the parameters that have an effect. The response-surface model is the predictive model, and every range claimed in PCR-009 will be derived from it.

3 Related documents

Table 3.1 lists the documents that scope this plan and the documents that depend on it. RA-001 sets the study scope, PCP-009 plans the study and PCR-009 reports it, and PCR-009 rolls up into PCMR-001.

Table 3.1: Related documents.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-009	Process Characterization Report (Small-Virus Retentive Filtration (Step 9))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

Table 3.2 lists the controlled procedures and validated analytical methods the study will be executed under. SOP-2012 governs operation of the step and the pre-use and post-use integrity tests, and the two method validations cover the infectivity assays for the model viruses.

Table 3.2: Controlled procedures and analytical methods for the study.

Refer- ence	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2012	Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation

Refer- ence	Title	Type
AMV- 3017	XMuLV Infectivity Titre (TCID50)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The step removes virus by size. The membrane carries a pore size distribution chosen to retain the smallest viruses of concern, the parvoviruses, while the antibody passes through. Retention is a sieving mechanism, so it depends on the integrity of the pore structure and on how the effective pore size of the membrane changes as it fouls over the course of a batch.

MVM sits close to the pore size cut-off, and its retention depends on the fraction of pores at the large end of the distribution and on the flow through them. As the membrane fouls with product and with residual impurity, flux redistributes toward the least fouled and largest pores, and small virus can pass. MVM breakthrough therefore rises with the volume filtered per unit area, and the highest load of a batch is the point at which retention is weakest. A low transmembrane pressure or an interruption of flow can also allow virus held within the pore structure to diffuse through, so pressure is controlled as well as volume.

XMuLV is many times larger than the pores. It is retained completely while the membrane is intact, and its clearance is insensitive to the operating parameters over any range that can be run. Membrane integrity is confirmed after each batch by the post-use integrity test under SOP-2012, and a membrane that passes that test has no pore large enough to pass a retrovirus.

Step yield is limited only by product held up in the fouled membrane and in the system, because the antibody is small enough to pass the pores freely. Yield is therefore expected to be high and to fall slowly as the volumetric load rises.

The step is a platform operation. It has been used to manufacture three previously licensed humanized IgG1 antibodies, X-Mab, Y-Mab and Z-Mab, and to supply A-Mab clinical material, with the same filter type and the same mode of operation (CMC Biotech Working Group 2009). The mechanism of removal is size exclusion and does not depend on the identity of the antibody, so the platform experience justifies the ranges chosen in §6.1 and supports the expectation that MVM retention falls with volumetric load while XMuLV retention does not. Prior knowledge does not remove the need for the study. The material presented to the filter is the anion exchange flow-through pool of the A-Mab process, and how that pool fouls the membrane is specific to the product, so the maximum load and the pressure range will be established on A-Mab material.

4.2 Quality attributes in scope

The step forms no quality attribute. It contributes log reduction to the two viral safety attributes in Table 4.1, both of which are cumulative requirements on the drug substance and are met by several steps together.

Table 4.1: Quality attributes the step contributes clearance to.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6-20 log10 (cumulative)	VH	20
Viral clearance — XMuLV (enveloped)	viral safety	16.7-30 log10 (cumulative)	VH	20

Both attributes are of very high criticality, and the risk they carry is patient safety rather than efficacy. MVM is the harder of the two for this step. The low pH inactivation step is not credited with any MVM clearance, because MVM is non-enveloped and low pH acts on the lipid envelope, so the cumulative MVM requirement rests on anion exchange chromatography and on this step alone. XMuLV is cleared by all three viral safety steps and by the chromatography steps that make no claim, so the cumulative XMuLV requirement is met with a wide margin.

The acceptance values in Table 4.1 are cumulative across the process and cannot be applied to one step. What this step must deliver is derived from them in §7.

Step yield is a process performance response. No drug substance attribute maps to it, so it will be measured and reported but judged against no quality criterion.

4.3 Risk-based prioritization of parameters

RA-001 ranked the parameters of every unit operation twice, once for the potential impact of the parameter on quality attributes and once for its potential to interact with other parameters (International Council for Harmonisation 2023b). Parameters scoring high on either ranking were assigned to multivariate study, parameters scoring lower were assigned to univariate study, and parameters supported by platform knowledge alone were assigned to neither.

For this step the assessment placed 2 parameters in the multivariate group and 0 in the univariate group. Filtration volume was ranked high because MVM retention falls as the volume filtered per unit area rises, once fouling has carried the flux onto the largest pores. Filtration pressure was ranked high because a loss of pressure or a stop in flow lets MVM already held in the pore structure diffuse forward into the filtrate, and because a rise in pressure drives more filtrate through the pores that remain open. Filtration volume decides how much of the membrane is still open and pressure decides how fast the filtrate crosses what is left, so the fall in MVM retention

with load is expected to be steeper at the top of the pressure range than at the bottom. The designs must be able to estimate that interaction.

No parameter of this step was left out of the study on the basis of prior knowledge alone. The risk assessment recorded that the platform claim for this step is a viral safety claim, and that a claim of that kind should be supported by product specific data even where the mechanism is well understood.

5 Materials and methods

5.1 Scale-down model and its qualification

The study will be run on a laboratory scale filtration system fitted with the same membrane type and the same membrane chemistry as the commercial step. The model is scaled on membrane area, so that the volumetric load in L/m², the transmembrane pressure and the flux per unit area are the same as at commercial scale. Load material will be matched because the product and the residual impurity the pool carries are what foul the membrane, and a cleaner pool fouls it more slowly. Filtration temperature will be matched because a higher temperature lowers the viscosity of the pool and raises the flux at a given pressure. The hold-up volume per unit area will be matched because it fixes how much product is left wetting the membrane and the tubing at the end of a run, which is what limits the step yield.

Qualification will follow SOP-1001. Triplicate runs of the scale-down system will be operated at the set-points in Table 6.1 on representative anion exchange pool, and the flux decay profile, the step yield and the post-use integrity test result will be compared with the corresponding manufacturing scale records. The model will be accepted where no statistically significant difference is found between the scales in yield or in flux decay at the end of the load.

Virus cannot be spiked into a manufacturing batch, so every clearance figure claimed for this step comes from the scale-down model (International Council for Harmonisation 2023a). The weight of the claim therefore rests on this qualification. The qualification will be repeated if the membrane type, the filtration temperature or the composition of the load material changes, and PCR-009 will state the qualification result before it reports any log reduction.

5.2 Operation

The load is the flow-through pool of the anion exchange step. Each run will be operated at constant transmembrane pressure to the volumetric load set by the design point, and the filtrate will be collected in fractions across the load.

Before each run the filter will be wetted and its integrity tested, and after each run the post-use integrity test will be performed, both under SOP-2012. A run whose post-use integrity test fails will be excluded from the analysis and recorded as a deviation, because a failed integrity test means the membrane was not intact and the log reduction measured across it says nothing about the step.

Each design point will be executed once, with both model viruses spiked into the load before filtration. Step yield will be determined from the same run by protein mass balance, from the ultraviolet absorbance of the load and of the pooled filtrate. Runs will be executed in randomized order within each design.

Filtration will be continuous. The claim made from this study applies to uninterrupted operation, because an interruption of flow allows virus held in the pore structure to diffuse through it. An interruption at commercial scale is a deviation and lies outside the operating region this study will define.

5.3 Analytical methods

Virus titre will be determined by the validated methods listed in Table 5.1. MVM will be titred by TCID50 with qPCR confirmation, and XMuLV by TCID50.

Table 5.1: Validated analytical methods for the viral clearance responses.

Reference	Title	Type
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation

The log reduction factor of a run is the base ten logarithm of the total virus in the spiked load divided by the total virus in the filtrate, with both totals taken over the volume actually filtered. Cytotoxicity and interference controls will be run for the load matrix and the filtrate matrix, and a spike recovery control will be run for each virus stock (International Council for Harmonisation 2023a).

Where no virus is detected in the filtrate, the titre will be reported at the limit of detection of the assay and the log reduction reported as a lower bound rather than as a point value. The infectivity assays carry their own variability, and the acceptance criteria in §7 allow 0.35 log₁₀ for it.

5.4 Sampling plan

Samples will be taken from the load before spiking, from the spiked load, and from the filtrate. The filtrate will be sampled as sequential fractions across the volumetric load and as a pool at the end of the run. Fractionating the filtrate allows the log reduction to be resolved against the volume filtered within a single run, which is the variable that degrades MVM retention as the membrane fouls.

Each spiked load sample and each filtrate sample will be assayed for both model viruses. The unspiked load and the unspiked filtrate pool will be assayed for protein concentration for the yield calculation. Samples may be held frozen before assay where the method validation supports it.

Every sample will be identified by run number and by the volumetric load at which it was taken, and the identity will be recorded at the time of sampling.

5.5 Statistical methods

Both factors will be coded to the interval -1 to +1 across the characterization ranges in Table 6.1, so that the estimated effects are comparable between the two factors and between the two designs. Effects will be reported as the change in the response over half the characterization range.

The screening design will be analysed with a model carrying both main effects and their interaction. The response-surface design will be analysed with the full quadratic model in both factors. Coefficients will be reported with their standard errors and p-values, and effects significant at $\alpha = 0.05$ will be treated as real.

Model adequacy will be judged from three things. The lack of fit will be tested against the pure error estimated from the centre point replicates. The residuals will be plotted against the predicted value and on a normal quantile plot. The coefficient of determination and the residual standard error will be reported for each response. The centre point replicates also give the reproducibility of each response, which will be reported as a coefficient of variation and used as the pure error term.

Analyses will be performed under SOP-1002. The analysis will be run from a script under version control with a fixed random seed, so that every number in PCR-009 can be regenerated from the raw data.

6 Study design

6.1 Factors, ranges and study type

Table 6.1 lists the parameters of the step, their set-points, the ranges that will be studied and the normal operating ranges.

Table 6.1: Parameters, characterization ranges and normal operating ranges.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Filtration volume (load)	L/m ²	90	50-140	50-105	multivariate
Filtration pressure	psi	13	8-30	10-20	multivariate

Both parameters will be studied multivariately. The characterization range for the filtration volume runs from the lower NOR edge to 140 L/m², which lies above the upper NOR edge of 105 L/m². The range was widened upward only, because MVM retention falls as the volumetric load rises and holds steady below the set-point. The edge of acceptable performance therefore lies above the NOR and not below it, and a range extended downward would consume runs where nothing changes.

The pressure range runs from 8 to 30 psi and is widened on both sides of the NOR of 10 to 20 psi. Low pressures are included because a low transmembrane pressure allows retained virus to diffuse through the pore structure. High pressures are included because they raise the flux and compress the fouling layer, so more filtrate passes through the largest open pores and MVM retention is expected to fall. Both edges are reachable by the commercial equipment, so a range that covers them supports the deviation assessment as well as the operating limits.

Table 6.2 gives the letters used for the two factors in the design matrices of Appendices A and B and in the effect tables of PCR-009.

Table 6.2: Coded factor legend.

Code	Factor	Unit	Studied in
A	Filtration volume (load)	L/m ²	screening + RSM
B	Filtration pressure	psi	screening + RSM

6.2 Screening design

The screening study will be a two-level factorial in both factors with centre point replicates, 7 runs in total. 4 runs are the factorial points at the corners of the characterization ranges and 3 are replicates of the centre point.

With 2 factors the full factorial is executed, so both main effects and the interaction are estimated free of aliasing. A two-level design carries no estimate of curvature, which is tested at the response-surface stage. The screening runs establish the size and direction of the two effects, and the reproducibility of the infectivity assays from the centre point replicates, before the response-surface runs are committed.

The planned coded settings are given in Appendix A.

6.3 Response-surface design

The response-surface study will be a face-centred central composite design in both factors, 12 runs in total. It is made up of 4 factorial points, 4 axial points and 4 centre point replicates.

The axial points sit on the edges of the characterization ranges rather than beyond them. A rotatable design would place them outside those edges, and settings outside the ranges in Table 6.1 are not settings the step may be claimed for, so the face-centred form is used. The design supports the full quadratic model, which allows the response to curve against the volumetric load. Curvature is expected there, because the flux redistribution that lets small virus through accelerates as the membrane fouls, and a model that could only fit a straight line would over-predict retention at the top of the range.

Four replicates of the centre point give the pure error term for the lack of fit test in §5.5 and an estimate of run to run reproducibility that does not depend on the model. The planned coded and natural unit settings are given in Appendix B.

6.4 Univariate assessment

No parameter of this step is assigned to univariate study. Both parameters in Table 6.1 are factors of the two designs above, and the remaining operating variables of the step are held at their set-points under SOP-2012 and are supported by platform experience.

Where the study finds an effect that the two designs cannot separate, a univariate confirmation run may be added. Any such run will be justified against the data that prompted it and reported in PCR-009 with the same acceptance criteria as the designed runs.

7 Acceptance and decision criteria

Three kinds of criterion apply to this study. One applies to the scale-down model, one to the fitted models, and one to the responses themselves.

The scale-down model criterion is stated in §5.1. The screening runs will not begin until it has been met, because a model that does not represent the commercial step cannot support a viral clearance claim from it.

A fitted response-surface model will be accepted as predictive where the lack of fit test is not significant at $\alpha = 0.05$ against the pure error from the centre point replicates, where the residuals show no trend against the predicted value and no material departure from normality, and where the residual standard error is small against the distance between the predicted response and its criterion. Where a model fails these checks, the affected response will be reported descriptively and no range will be claimed from it.

Table 7.1 gives the criterion each response will be judged against.

Table 7.1: Acceptance criteria for the study responses.

Response	Criterion	Basis
MVM LRF ($\log_{10}$)	≥ 4.97	in-process
XMuLV LRF ($\log_{10}$)	≥ 5.47	in-process
Step yield	none	process performance

The two viral clearance responses are judged against an in-process criterion rather than against the drug substance requirement. The requirement in Table 4.1 is cumulative, at 8.6 $\log_{10}$ for MVM and 16.7 $\log_{10}$ for XMuLV. Subtracting the clearance credited to the other steps of the train leaves this step a floor of 3.89 $\log_{10}$ for MVM and 3.65 $\log_{10}$ for XMuLV. The criteria in Table 7.1 are tighter than those floors, because each is set from the log reduction credited to this step in the modular clearance strategy, less the assay allowance of 0.35 $\log_{10}$ described in §5.3. The criteria in Table 7.1 are the tighter of the two, so the study will be judged against them, and PCR-009 will quote the floors only to show the margin the cumulative claim retains.

Step yield carries no criterion. It will be reported with the operating region and used to show what the region costs in product, and no acceptance decision will rest on it.

An operating range for a parameter will be declared where the response-surface model predicts that both viral clearance responses meet their criteria across that range. The design space for the step is the part of the characterized region where both viral clearance responses meet their criteria at the same time, and PCR-009 will report

its size from the fitted models rather than assert that it is the whole characterized region.

Each parameter will be classified under SOP-4001 from the outcome of the study. A parameter whose variability affects a critical quality attribute will be classified as a CPP, or as a WC-CPP where the risk of operating outside the design space is low given the control capability of the commercial equipment. A parameter that affects process performance but no quality attribute will be classified as a KPP, and a parameter with neither effect as a GPP (International Council for Harmonisation 2009).

The study will be considered successful where the scale-down model is qualified, both viral clearance models are accepted as predictive, and a proven acceptable range containing the NOR is established for each parameter against each of the two viral clearance responses. Where no such range exists, PCR-009 will report the range that does exist and the operating limit that follows from it.

8 Proven acceptable ranges (planned analysis)

A proven acceptable range is the range of a parameter over which the response meets its criterion, given a stated treatment of the other parameters (International Council for Harmonisation 2009). Two analyses will be run for each parameter and each of the two viral clearance responses, and both will be reported.

The first holds the other parameter at its set-point and scans the parameter of interest across its characterization range on a grid of 201 points, using the fitted response-surface model. It answers what the parameter may do while the rest of the step is operated at target.

The second varies the other parameter within its NOR by Monte-Carlo simulation of the fitted model, with 2,000 draws at each of 81 grid points, and takes the range over which the 95 % prediction interval of the response stays within acceptance. The draws carry the residual variability of the model as well as the variability of the other parameter, so this range will normally be the narrower of the two. Where the two analyses disagree, the range from the second is the one carried into the control strategy, because the commercial process holds each parameter somewhere within its normal operating range and not at its set-point.

The reported range in each case is the contiguous interval that contains the set-point. Where the predicted response leaves acceptance inside the characterization range, the proven acceptable range is narrower than the range studied, and PCR-009 will state the value at which the response crosses its criterion.

Each proven acceptable range will be reported with a plot of the response against the parameter, with the acceptance limit drawn, the acceptable region shaded green, and the set-point and the NOR marked. The two analyses will be shown as the two panels of one figure, so that the cost of the robustness criterion is visible.

The ranges themselves will be reported in PCR-009.

9 Data management and integrity

Raw data will be recorded at the time of the activity, in the electronic laboratory notebook for the process runs and in the laboratory information management system for the analytical results. Records will meet the ALCOA+ expectations, and the audit trails of both systems will be retained for the life of the application (International Council for Harmonisation 2008).

Analytical results will be transferred to the analysis by controlled file export. The fitted models, the tables and the figures of PCR-009 will be generated from the exported data by a single reproducible run of the analysis script, and a second reviewer will confirm that the reported numbers come from that run. Any manual entry of a value into the analysis is a data integrity finding and will be handled as one.

Deviations from this plan will be recorded when they occur and reported in PCR-009 with their assessed impact on the study. A deviation that invalidates a design point will be dispositioned before the affected runs are analysed, and where the disposition is to repeat the affected runs, the superseded data will be retained and referenced rather than deleted.

10 Roles and responsibilities

Table 10.1 gives the functions involved in the study and what each is accountable for. The named individuals are recorded in the approval table of each document rather than here, so that a change of personnel does not require a revision of this plan.

Table 10.1: Functions and responsibilities for the study.

Function	Responsibility
Process Development / MSAT	Owns this plan, executes the filtration runs, authors PCR-009
Analytical Development	Performs the infectivity assays and the protein determinations under the validated methods
Statistics	Designs the experiments, performs the analysis, reviews model adequacy
Manufacturing Science, receiving site	Reviews the operating ranges against the control capability of the commercial equipment
Quality Assurance	Approves this plan and PCR-009, and dispositions deviations

The receiving site reviews the ranges before they are fixed, because a range the commercial equipment cannot hold is of no use in the control strategy (International Council for Harmonisation 2008).

11 Deliverables and schedule

The deliverable of the study is PCR-009. It will report the qualification of the scale-down model, the screening effects, the response-surface models with their diagnostics, the proven acceptable ranges, the design space of the step, the parameter classification and the contribution of the step to the control strategy and to the modular viral clearance claim.

The study will be executed in four phases, in this order.

1. Qualification of the scale-down model, against the criterion in §5.1.
2. The screening runs and their analysis.
3. The response-surface runs and their analysis.
4. The proven acceptable range analysis, the classification and the report.

Each phase begins when the preceding phase has met its criterion in §7. No calendar dates are fixed in this plan. The study is scheduled against the transfer programme in PTP-001, and its report rolls up into PCMR-001 with the reports of the other two viral safety steps.

12 Risks and assumptions

The clearance claim for this step rests entirely on a scale-down model, because virus cannot be spiked at manufacturing scale. If the qualification in §5.1 fails, the study cannot proceed. The mitigation is to re-scale the model on membrane area and on hold-up volume and to repeat the qualification, and any change made to the model will be recorded in PCR-009.

Spiking adds material to the load, and retention depends on how the membrane fouls. A spike that fouls the membrane differently from unspiked process material would bias the measured log reduction, and the bias would be largest at the top of the volumetric range, which is the part of the range the claim depends on. The virus stock will therefore be clarified before addition and its volume kept small against the load volume, and the flux decay of the spiked runs will be compared with that of the unspiked qualification runs.

The log reduction that can be measured is bounded by the titre of the spiked load and by the limit of detection of the infectivity assay. Where no virus is detected in the filtrate, the result is a lower bound, and a lower bound that falls short of the criterion in §7 cannot be resolved by repeating the assay. The load will therefore be spiked to a titre that allows the criterion to be demonstrated with margin, and the achieved titre will be reported with each run.

The study assumes that the anion exchange pool used in the runs represents commercial material. Anion exchange is the step immediately upstream and its own characterization is reported in PCR-008. Pool from a run that was itself out of trend will not be used, and the load material used in each filtration run will be identified in PCR-009.

The study assumes uninterrupted filtration, as stated in §5.2. The ranges it produces do not cover an interruption of flow, and an interruption at commercial scale will be assessed as a deviation against the mechanism in §4.1 rather than against these ranges.

Material supply is the principal risk to the schedule. Each design point consumes anion exchange pool, and the runs will be scheduled against pool availability rather than against a fixed calendar.

13 Approvals

This plan takes effect when it has been approved by the functions listed in Table 1. Approval records that the study design in §6, the acceptance criteria in §7 and the data management provisions in §9 are adequate to support the transfer and the viral clearance claim.

A change to the factors, to the ranges in Table 6.1 or to the acceptance criteria in Table 7.1 after approval requires a revision of this plan and re-approval before the affected runs are executed. A change made after the affected runs have been executed will be reported in PCR-009 as a deviation from this plan.

14 Appendix A — Planned screening design matrix

Table 14.1 gives the coded settings of the 7 screening runs. The letters are those of Table 6.2, and the coded levels -1, 0 and +1 correspond to the low edge, the midpoint and the high edge of each characterization range in Table 6.1.

Table 14.1: Planned screening design, coded settings.

Run	Type	A	B
1	factorial	-1	-1
2	factorial	-1	1
3	factorial	1	-1
4	factorial	1	1
5	center	0	0
6	center	0	0
7	center	0	0

Table 14.2 gives the same design in the units the runs will be executed in.

Table 14.2: Planned screening design, natural unit settings.

Run	Type	Filtration volume (load)	Filtration pressure
1	factorial	50	8
2	factorial	50	30
3	factorial	140	8
4	factorial	140	30
5	center	95	19
6	center	95	19
7	center	95	19

The executed matrices with their measured responses will appear in the appendices of PCR-009.

15 Appendix B — Planned response-surface design matrix

Table 15.1 gives the coded settings of the 12 response-surface runs, in the same coding as Appendix A. The axial runs carry one factor at an edge of its characterization range and the other at the midpoint, and the centre point is replicated 4 times.

Table 15.1: Planned response-surface design, coded settings.

Run	Type	A	B
1	factorial	-1	-1
2	factorial	-1	1
3	factorial	1	-1
4	factorial	1	1
5	axial	-1	0
6	axial	1	0
7	axial	0	-1
8	axial	0	1
9	center	0	0
10	center	0	0
11	center	0	0
12	center	0	0

Table 15.2 gives the same design in natural units.

Table 15.2: Planned response-surface design, natural unit settings.

Run	Type	Filtration volume (load)	Filtration pressure
1	factorial	50	8
2	factorial	50	30
3	factorial	140	8
4	factorial	140	30
5	axial	50	19
6	axial	140	19
7	axial	95	8
8	axial	95	30
9	center	95	19
10	center	95	19
11	center	95	19

Run	Type	Filtration volume (load)	Filtration pressure
12	center	95	19

The execution order within each design will be randomized, so the run numbers in these tables identify the design point and not the sequence in which it will be run.

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